

Density and Mobility-driven Evaluation of Broadcast Algorithms for MANETs

Raziel Carvajal Gómez¹, Inti Gonzalez-Herrera², Yérom-David Bromberg³, Laurent Réveillère², Etienne Rivière¹
1. University of Neuchâtel, Switzerland, 2. LaBRI/University of Bordeaux, France, 3. IRISA/University of Rennes 1, France.

etienne.riviere@unine.ch

Abstract—Broadcast is a fundamental operation in Mobile Ad-Hoc Networks (MANETs). A large variety of broadcast algorithms have been proposed. They differ in the way message forwarding between nodes is controlled in time and space, and in the level of information about the topology that this control requires. Deployment scenarios for MANETs vary widely, in particular in terms of nodes density and mobility. The choice of a broadcast algorithm depends on its expected reachability and energy cost, which are both highly impacted by the deployment context. In this work, we are interested in overcoming the lack of comprehensive comparison of the costs and effectiveness of broadcast algorithms for MANETs, and ease an informed choice between them based on target environmental conditions. We describe the results of an independent experimental study of five algorithms, representative of the principal design alternatives. Our study reveals that the best algorithm for a given situation, such as a high density and a stable network, is not necessarily the most appropriate for a different situation such as a sparse and mobile network. We identify the algorithms characteristics that are correlated with these differences and discuss the pros and cons of each designs. Our experimental framework and results are available in the open for use by the community.

Index Terms—Broadcast, MANETs, Experimentation

I. INTRODUCTION

With the rise of the Internet of Things (IoT), the number and diversity of connected devices is on the rise. Leaders of the industry, such as Cisco, predict that 50 billion devices will be connected together by 2020 [1]. The goal of the IoT is to link this diversity of objects and promote machine-to-machine interactions. These interactions enable the capture and/or distribution of data that can be turned into information and feedback delivered to end users.

The advent of the IoT has led to a strong revival of interest in mobile ad-hoc networks (MANETs). A MANET is a mobile network that does not rely on a fixed infrastructure. Instead, it is a self-configuring network where nodes connect directly to each other, and where there is no guarantee for end-to-end connectivity. The topology of a MANET changes frequently due to the mobility of participants. Nodes are connected using a wireless medium, there is no central entity in charge of routing, and most participants operate on battery-powered devices with limited processing power and bandwidth. Applications based on MANETs [2] range from sensor networks to end-user mobile applications, such as traffic jam prevention system, information dissemination in crowds, data gathering in hostile environments, and peer-to-peer mobile games.

Many network functions in MANETs, such as routing for instance, rely on the fundamental *broadcast operation* that consists in delivering a message from a source node to all other nodes in the network. The broadcast problem for MANETs has received considerable attention. In particular, existing solutions attempt to:

- *Maximize reachability.* Percentage of nodes that receive a message. Basically, broadcast messages must reach every node of a network.
- *Minimize energy consumption and bandwidth usage.* Generally, the efficiency of an algorithm is measured by the amount of energy, processing power and network bandwidth it uses to disseminate messages.
- *Be mobility-tolerant and density aware.* A broadcast algorithm should guarantee high reachability and cope with mobility patterns under a large variety of deployment scenarios, where the connectivity between nodes is changing constantly.

Many algorithms have been proposed to solve the broadcast problem in MANETs [3]–[8]. All these algorithms broadcast by using a form of *controlled flooding*: since the transmission range of a source node is typically much smaller than the network area, broadcasting is achieved by message retransmission between nodes to cover the entire network. Broadcast algorithms for MANETs mostly differ in the way they control this forwarding operation. Particularly, the control happens in space and time by deciding *which* nodes should forward a message and *when* this operation takes place. Algorithms can also be distinguished on their use of, or independence from, knowledge about the network topology, nodes locations, and network status.

Besides these differences in the nature of messages forwarding, broadcast algorithms are also largely influenced by external factors such as the density of the network, and the mobility rate of its nodes. The performance of broadcast protocols has been studied individually [9], [10] and extensive work have been done on their classification taking into account the way they work [11], [12]. However, to the best of our knowledge, there is a lack of comprehensive experimental comparison of existing solutions allowing an understanding under various deployment scenarios. Moreover, there is not a common set of *broadcasting metrics* follow by the network community to overview the assets of solutions one can find at the literature. **Contributions.** Our goal in this paper is to explore the relative performance and costs of representative MANET broadcast

algorithms in a variety of deployment conditions, in order to ease the task of selecting the most appropriate algorithm for a given situation. In addition to the *simple flooding* algorithm, our study considers four algorithms that are representative of their class:

- *ABBA* controls the dissemination process in time, by setting up timers prior to dissemination that are inversely proportional to the probability that the same message will be received from other neighbors [3].
- *MPR* maintains an overlay structure to control the broadcast process. This overlay is built using a heuristic that aims at minimizing the number of forwarding nodes [4].
- The *node-degree* approach builds a source-independent approximation of the minimum Connected Dominating Set (CDS). This overlay allows controlling which nodes should propagate incoming messages [5].
- *ProbFlood*, a probabilistic approach where the likelihood of transmitting a message depends on the local density around each node [6].

The contributions of our performance comparison study aim at helping practitioners to decide which is the most appropriate algorithm for a certain deployment scenario. Firstly, our results show that there is no one-size-fits-all to broadcast in MANETs. Secondly, we identify which are the algorithms that better match the studied deployment scenarios. These findings can further be used to design a broadcast approach that changes its behavior at runtime based on contextual information. Finally, we outline an adaptive broadcast mechanism in which the algorithm to use is always the best fit for a given deployment scenario.

Outline. The rest of this paper is organized as follows. Section II presents the algorithms we evaluate in our experiments. In Section III we survey related work on the evaluation of broadcast algorithms for MANETs and discuss its limitations. We present our experimental setup in Section IV and our evaluation results in Section V. Section VI summarizes the main findings of this paper. Finally, we discuss the perspective of this work in Section VII and conclude in Section VIII.

II. BACKGROUND

A straightforward approach to broadcast is *blind flooding*. When one host receives a broadcast message m for the first time, it forwards m to its neighbors; the reception of further copies of m are ignored. This approach suffers from a number of drawbacks known as *the broadcast storm problem* [13]. Given that nodes forward each message at least once, the communication medium rapidly becomes saturated, the likelihood of collisions increases and message loss as well.

Generally, the design of broadcast algorithms for MANETs is a trade-off between two goals: (i) guaranteeing a suitable level of reachability, and (ii) optimize resources usage of nodes, prior to and during the actual dissemination. As a consequence, broadcast algorithms attempt to reduce the number of forwarding nodes to the minimum required to keep 100% of reachability; avoiding the depletion of the energy in nodes' batteries to improve the overall life time

of communication. In practice, however, it is hard to decide which nodes should retransmit a message.

Naturally, the lack of a central authority in MANETs implies that broadcast algorithms must be fully distributed and rely only on *local* data available at each node. One can find many solutions for efficient message dissemination at the literature. Ruiz and Bouvry [11] propose a survey of broadcast algorithms for MANETs. The authors define a taxonomy to classify current state-of-the-art algorithms based the way they control the forwarding operation. In this study, we have selected five algorithms, including simple flooding, that are representative of their class. We consider only protocols in which the transmission range of the antenna is fixed. Variable-range broadcast protocols are kept as future work.

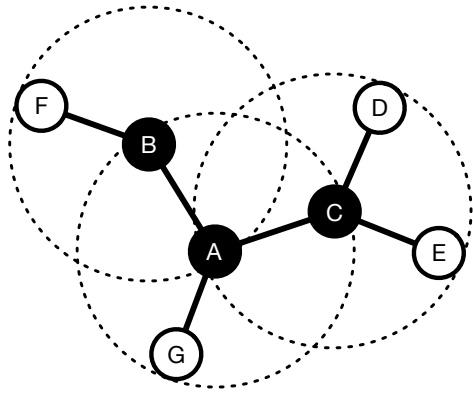
Broadcast algorithms can be first classified in two main classes, depending on whether or not they maintain an underlying virtual topology between nodes.

A. Topology-based algorithms

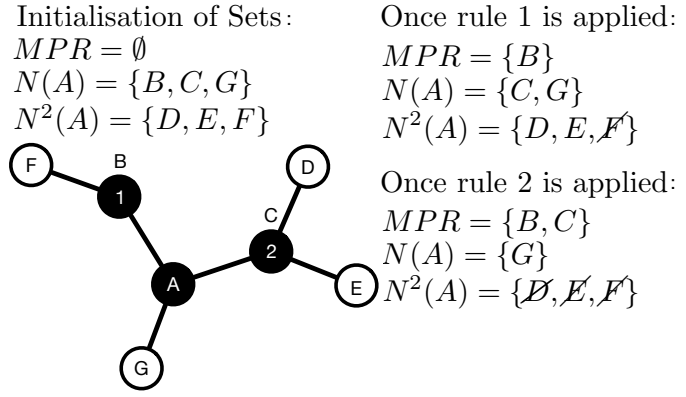
One way to broadcast is through the use of an underlying topology. A backbone or spine is a subset of nodes where its members are allowed to forward messages due to their position in space. Any other node out of the backbone acts as message receiver. For instance, Figure 1(a) depicts a small network of nodes where node A starts with the message dissemination. If only nodes B and C perform the retransmission of any incoming message, the reachability of this network will be 100%. Of course, the complexity of a topology based algorithm remains in the construction of this backbone. Connected dominating set (CDS) is the formal term for such a backbone [5], formally talking, nodes are partitioned in such a way that they either belong to the CDS or are adjacent to any other node at it. Additionally, connectivity is another property of a graph containing a CDS. In this study, we consider two topology-based algorithms *Multipoint relaying (MPR)* and one *CDS-based technique*, which require a permanent exchange of control messages (aka *HELLO* messages) among nodes to reduce the number retransmissions.

a) *MPR*: The CDS obtained by the technique propose in [4] is called a MPR set. Nodes start with an empty *MPR* set knowing their one-hop and two-hop neighbors, represented by the sets N and N^2 respectively, thanks to previous exchanges of control messages. The procedure to find relay nodes is two-fold. Firstly, any one-hop neighbor $n \in N$ is chosen as relay node of the MPR set if n is a unique neighbor of some other node at N^2 . Secondly, while the current MPR set do not cover every node in N^2 , we iterate over each remaining neighbor $n \in N$ and we count how many two-hop neighbors it covers. n will be a relay node only if the number of covered two-hop neighbors is maximal.

Figure 1(b) shows the steps taken by node A to compute its *MPR* set. Once the data about one-hop and two-hop neighbors is exchanged, after applying rule No 1, node B is considered as relay node because it is a unique neighbor of F . After applying rule No 2, node C is included at *MPR* given that



(a) Backbone of 3 nodes shown by the black filled-in circles.



(b) Steps to get the MPR set of peer A

Fig. 1: Virtual topology to reach 100% reachability in one MANET.

the number of covered two-hop neighbors (i. e. two) is bigger than those neighbors cover by G (which is zero).

b) CDS-based technique: The decentralize algorithm in [5], firstly approximates one CDS via a marking procedure and secondly, the number of relays nodes is minimized. As in MPR, data about one-hop and two-hop neighbors is required as well. The marking procedure simply identifies whether one node n has any pair of neighbors that are unconnected, n marks itself as a relay node if this condition takes place. For instance, in Figure 1(a) node A marks itself as relay node because nodes B and G are unconnected. To reduce the elements of a CDS obtained by the marking procedure, authors in [5] take into consideration either *the degree of nodes* or *the remaining energy in nodes' batteries*. We chose the former approach to be fair in the comparison with MPR.

The node-degree-based rules are basically operations of first-order logic between the one-hop and two-hop sets of nodes' neighbors. Due to the lack of space we limit our explanation saying that our current implementation of the CDS technique follows rules 2 and 2a. One can refer to [5] for having more details about these two rules.

B. Context-aware and context-oblivious algorithms

Using a virtual topology implies an overload on the communication medium due to the constant exchange of control messages. When nodes move, the frequency on sending these messages must be increased to keep a virtual topology with the correct set of relay nodes. This results in a higher probability of collisions. Moreover, our study considers a second category of broadcast protocols that do not require the exchange of control messages to build a virtual topology, instead, data is piggybacked in messages before their retransmission. This data could be either context-oblivious or context-aware. The former forwards messages with a predefine probability while the latter use knowledge about the current state of the network (for example, nodes' density).

a) Probabilistic Flooding: This is a context-aware and context-oblivious approach [6] that uses information about one-hop neighbors and a probability to perform retransmissions. Nodes retransmit messages based on the distance among neighbors and a probabilistic scheme. This differs from the canonical probabilistic approach where every node in the network has the same probability of retransmitting a message regardless of the density. The probability of retransmitting a message decreases as the local density increases.

b) ABBA: Area-based beacon-less protocol [3]. The flooding process in ABBA is controlled in both space and time. This algorithm requires nodes to know their physical location in addition to the transmission range of nodes' radio. When one broadcast message M is received, nodes compute the angle α covered by this reception. Note that α is paired to a unique message. M is scheduled for being retransmitted after T seconds, this timer is computed as follows $T = T_{max} - T_{max} \frac{\alpha}{2\pi}$, where T_{max} is a fixed upper bound that represents the maximum waiting time. Prior of a new reception of M before T seconds, nodes cancel the current timer and schedule a new one. In other words, the bigger the angle covered by message receptions, the shorter the time to wait before performing retransmissions. With one exception, message retransmissions are canceled if α reaches a value of 2π .

The diversity of algorithms to broadcast in MANETs is vast and choosing one approach to cope with a particular deployment scenario is a challenging task. While some evaluations have been proposed to compare existing approaches, they usually target fixed scenarios; this avoids its use in other contexts. In this work, our goal is to define an experimental framework to fairly assess representative algorithms to broadcast in MANETs. In the next section, we discuss previous works one can find in the literature.

III. RELATED WORK

Many previous studies evaluate the performance of broadcast algorithms for MANETs. These studies are of two types:

proposal and evaluation of new broadcast algorithms (NPE), and fully dedicated performance comparison studies (CS). In this section, we discuss the evaluation criteria and the results obtained in a representative set of previous works.

In Table I we summarize the related work. First, in column *multi-protocol* we analyze whether a study compares several broadcast protocols. Next, we look at how these studies compare the evaluation of these protocols. That is, column *multi-class* indicates if a study only compares protocols in the same class instead of comparing many classes. Since the network density is extremely important in the context of wireless networks, we highlight in column *multiple densities* what studies consider the impact of this parameter. The next two evaluation criteria we consider are related to two characteristics of MANETs – mobility and energy consumption. We show in Table I whether a specific study takes into account the influence of mobility on the overall performance of the protocol and whether the impact on energy consumption is evaluated. Finally, the last column indicates whether the condition of the network usage are taken into consideration. We believe this last criterion is important because in a wireless network, in which there are many collisions, the performance of a broadcast protocol varies depending on the network load. In the remainder of this Section we discuss further details of how broadcast algorithms are evaluated in previous works.

In their foundational work on the topic, Ni et al. [13] identify the drawbacks of using basic flooding schemes, such as blind and simple flooding, to perform broadcast. To avoid the main problems, the authors propose different algorithms that, by construction, outperform simple flooding in a set of metrics. Since by construction such approaches are an improvement, the authors then proceed to evaluate them without considering simple flooding. This is a common pattern in evaluating broadcast algorithms. For instance, column *multi-protocol* in Table I shows that although most studies compare many protocols, many of them do not consider simple flooding in their evaluation. This form of evaluation is nevertheless limited because, as shown in [9], comparing against simple flooding is important when metrics such as energy consumption and mobility are taken into consideration.

Interestingly, most studies only consider protocols in the same class. For instance, in [7], the authors only compare algorithms with *fixed transmission range* that build a Connected Dominating Set (CDS) as their *underlying topology*. Likewise, only the construction of *cluster-based underlying topologies* is evaluated in [16]. Meanwhile, in [6] only *probabilistic* approaches with *context awareness* are considered. Finally, in [14] the authors only compare algorithms that build a *tree* by varying the transmission range. That being said, there are indeed studies that compare protocols from different classes. Previous work compare *probabilistic* and *area-based* algorithms [10], [13]; while some studies evaluate the construction of *CDS-based* and *cluster-based* topologies to broadcast [8], [15]. Finally, we also find previous work comparing protocols from diverse classes [9]. Since most previous studies don't consider multiple classes or only take into account few classes of

algorithms, we think that this is one of the main limitations in the related work.

In most previous work, it is assumed that the overall energy consumption decreases when fewer nodes rebroadcast messages. As a consequence, many studies only focus on measuring the improvement in terms of this latter metric instead of looking at the real energy consumption. However, it is worth mentioning that such an approach has disadvantages because the cost of building and maintaining an underlying topology is not taken into consideration. Despite of this, Table I shows that most studies do not directly evaluate energy consumption. Other criteria such as mobility and the influence of the network load on the performance of broadcast algorithms are barely evaluated. In particular, this latter criterion is only evaluated by adding a probability of dropping packages [4], [13] to the MAC layer. Although this is a valid initial approximation, we think it is possible to better measure the effect of network load considering more realistic models of the MAC layer.

IV. EXPERIMENTAL SCENARIO

Our experiments are inspired by a real life scenario: a physical and spontaneous *demonstration* gathering where the infrastructure for mobile communication is not able to serve the amount of clients and where individuals use their mobile phones in an ad-hoc mode to communicate. In this scenario, the efficient dissemination of some messages to all participants is critical. To simulate this kind of scenario, we must consider two parameters that influence the performance of broadcast in MANETs: *network density* and *nodes mobility*. In the following, we describe the simulation environment and the parameters used in our study.

Simulation tool

For our performance comparison study, we rely on INET3.3/OMNET++4.6, a simulation framework specifically aimed at evaluating networking algorithms [?]. This tool provides models for many protocols of the network stack such as 802.11g wireless communication [17]. In addition, it contains models to describe and deal with mobility and energy consumption aspects.

Communication Scenario

In a *demonstration* scenario, some attendees have mobile devices with IEEE 802.11 capability. Participants use such devices to broadcast messages using a MANET and they might attempt to do so at nearly the same time. The messages to broadcast are usually *short* because most of them are generated using the input capabilities of mobile devices.

To simulate these conditions, we set the number of broadcast messages to 100 and the interval between messages to 0.5 seconds in our experiments. In addition, the source of each message is randomly selected during the generation of the network (see next subsection). Finally, the content of each message is randomly generated, unique and with a length of 32 bytes at most.

Study	Type of Study	Multiple protocols	Uses simple flooding	Multiple classes	Many densities	Mobility	Energy Consumption	Different probability of error/network load
Comparative Performance Study [9]	CS	✓	✓	✓	✓	✓	×	×
Context-aware broadcasting [10]	CS	✓	✓	✓	✓	✓	×	×
Broadcast Storm Problem [13]	NPE	✓	×	✓	×	×	×	✓
EEMPR [7]	NPE	✓	×	×	✓	×	×	×
Neighbor Elimination Scheme [8]	NPE	✓	×	✓	✓	✓	×	×
EWMA [14]	NPE	✓	×	×	✓	×	✓	×
FNSB [15]	NPE	✓	×	✓	✓	✓	×	×
K-HOP [16]	NPE	✓	×	×	✓	×	×	×
MPR [4]	NPE	×	✓	×	×	×	×	✓
Border Node Retransmission [6]	NPE	✓	×	×	✓	×	×	×
ABBA [3]	NPE	✓	×	×	✓	×	×	✓

TABLE I: Survey of evaluation criteria used in different performance studies. In the second column, CS stands for *comparison study* and NPE for *New Protocol Evaluation*

Node Density

We are interested in generating scenarios with different node density, this allows us to simulate more accurately the *demonstration* use case. To characterize the node density we use three parameters: transmission range, map size and number of nodes. As shown in [18], there is a dependency between these parameters that allows us to build network topologies with a given density by using random geometric graphs.

In our experiments, all nodes have a transmission range of 20 meters.¹ Meanwhile the number of nodes is fixed to 500, the mode reported by previous studies on the size of demonstrations in Western Europe [?]. To create a network with a given node density, we use the values of transmission range and number of node to compute the map size using the aforementioned dependency relationship. Then, we generate a laying on a square map by using a procedure based on random geometric graphs. The output of this procedure is a network topology with the desired density.

Table II summarizes the simulations parameters we use to configure the density. We instantiate 8 networks of 500 nodes each and set the densities to 5,10,15,20,25,30,35 and 40, a range of values that is frequently found in the literature.

Parameter	Setting
Number of Nodes	500
Transmission Range	20 m
Densities	5–40
Map Size for smallest density	354 m x 354 m
Map Size for largest density	125 m x 125 m

TABLE II: Simulations parameters affecting node density.

¹Theoretically, IEEE 802.11 devices allow communication in a range between 20 and 140 meters depending on the line of sight. Previous empirical tests show that it can effectively be between 20 and 30 m in indoor or crowded places [10].

Mobility

In our evaluation, we use the *truncated-levy walk* (LW) mobility model, as it characterizes the mobility traces of humans carrying wireless devices [19]. This model generates traces composed of frequent short walks and occasional long rides to distant locations. In our experiments, we define two kind of scenarios: one without mobility (*static*) and another with mobility at walking speed (*dynamic*). The generation of *static* scenarios was presented in the previous subsection. To generate *dynamic* scenarios, we begin with a *static one*. Then, we use the LW model to generate mobility traces for each node. The parameters of the LW model are set in such a way that the average node speed is close to 1.4 m/s, a value reported in the literature [20] as an estimation of the preferred walking speed. To do so, we use the following values to feed the mobility model:

Parameter	Value	Description
FL_EXP	-3.9	Exponent of the flight length distribution
FL_MAX	50	Maximum value of the flight length distribution
WT_EXP	-1.8	Exponent of the waiting time distribution
WT_MAX	100	Maximum value of the waiting time distribution

TABLE III: Configuration of the LW mobility model.

Afterwards, we perform several checks on the traces to guarantee that: i) the average speeds are indeed close to 1.4 m/s, ii) the node density is the desired one during all the simulation time, and iii) the topology remains connected all the time.² Should any of these checks fail, we discard the mobility traces.

²Except for a very low density (5) where we are unable to generate a topology that remained connected during all the experiments.

Energy Consumption

The efficiency in energy consumption must be considered in evaluating protocols that use wireless communication. Since power consumption highly depends on the Wi-Fi chip used by mobile devices, we chose to model power consumption using the specifications of an actual chip found in modern mobile devices. The Broadcom BCM4329 [21] can be found in many modern devices, it supports the Wi-Fi 802.11g specification on the 2.4 GHz frequency band. Table IV shows the amount of energy that this chip requires in its different operation modes. This information is used in setting the power consumption model of INET.

Mode	Consumption
Sleep	648 nW
Receiver Listen	244.8 mW
Receiver Active (RX)	295.2 mW
Transmitter Active (TX)	1206 mW

TABLE IV: Power consumption per mode of the Broadcom BCM4329 Wi-Fi chip.

To simulate this chip, we use the *StateBasedEnergyConsumer* model provided by INET. This model measures how many Joules were consumed based on the time the chip spends on each mode (as described in Table IV). Formally, this energy consumption model uses three functions: $c : M \mapsto \mathbb{R}$, $r_m : \mathbb{R} \mapsto M$ and $t_m : \mathbb{R} \mapsto M$ where M is the set of operation modes of the chip, c is the consumption of the chip on each mode, and functions r_m, t_m map from simulation time to operation modes of the *receiver* and *transceiver*, respectively. INET then computes the energy consumption $C(t)$ using: $C(t) = \int_0^{t_{\text{simulation}}} (r_m \circ c)(t) + (t_m \circ c)(t) dt$. The function m is given and INET computes both r_m and t_m during the simulation.

Algorithm specific parameters

The behavior of each protocol depends on the values of its parameters. To configure these parameters meaningfully, we choose values previously reported in the literature. For instance, we use 0.5 seconds as the *hello time* for *MPR*, *ProbFlood* and *node-degree-based CDS*. In the case of *ABBA*, we use 0.3 seconds as the maximum waiting time. Finally, for *ProbFlood* we assign the values $k = 15$ and $\sigma = 1$.

Simulation time

As mentioned in Section II, some algorithms build a topology to drive the broadcast operation. To guarantee that the first broadcast message is properly handled by all protocols, we set an initial warm-up period of 5 seconds before the first message is broadcast. The warm-up phase can be used, for instance, to create the initial underlying topology in CDS-based algorithms. Likewise, in our evaluation, we stop each experiment a fixed amount of time (5 seconds) after the beginning of the last broadcast operation. This grace period allows the last message to reach all nodes. The total simulation time is then $10 + \delta t \times M$, where δt is the interval between broadcast messages and M is the number of scheduled messages.

V. EVALUATION

In the previous section, we describe the deployment scenarios we use to evaluate the broadcast algorithms. In this section, we present the results of our experiments. Such results are expressed in terms of the following metrics:

▷ **Saved Rebroadcasts (SRE)** – number of retransmissions we avoid by using an algorithm A instead of *simple flooding*, which is considered the worst case. To compute the SRE of one algorithm A in a network of N nodes, we measure the number R_m of retransmissions made by A to disseminate a message m . The SRE_m for message m is then $N - R_m$. Since in our experiments we are disseminating many messages, we simply compute the mean SRE and use it as metric for our comparison study. This metric is useful to assess the performance of algorithms in terms of both network and energy usage.

▷ **Duplicate messages** – number of times one broadcast message is received after its first reception.

▷ **Reachability** – percentage of nodes that successfully receive one broadcast message at least once.

▷ **Broadcast time** – time to disseminate one message with a reachability of 100%.

▷ **Energy consumption** – energy consumed by nodes during all broadcast session.

We analyze every metric with and without mobility over different number of neighbors (range of node densities) as refereed at Section IV.

A. Network overload

The proportion of SRE and duplicate messages determine to which extent the communication medium is saturated. Figure 2(a) depicts the SRE in static scenarios. As can be seen, MPR outperforms all other protocols. However, the difference between MPR and all other algorithms decreases as the density increases. Interestingly, ABBA reduces the number of retransmissions more efficiently than the node-degree technique for densities above 10 but it does not surpass MPR. The competitiveness of ABBA was expected to increase with the density because ABBA can cancel more retransmissions, but it is surprising to see it surpassing node-degree by 10%.

In dynamic scenarios, as shown in Figure 2(c), we observe how node mobility has a negative impact on the performance of topology-based algorithms, particularly on MPR. Indeed, MPR and node-degree protocols keep nodes' neighbors information that is soon out of date due to the constant changes. Additionally, collisions of control messages in these algorithms is another explanation for the bad performance. While ABBA maintains its robustness when nodes move, it must be said that this algorithm requires a GPS in its functioning; the impact of such a service is not considered in this paper.

The proportion of duplicate messages per algorithm is shown in Figures 2(b) and 2(d). As this value varies per each broadcast session, we are using *boxplots* to present a distribution of the metric. One can notice that the number of duplicate messages is never equal to the density of nodes because collisions provoke the loss of broadcast messages. For instance, with density 30

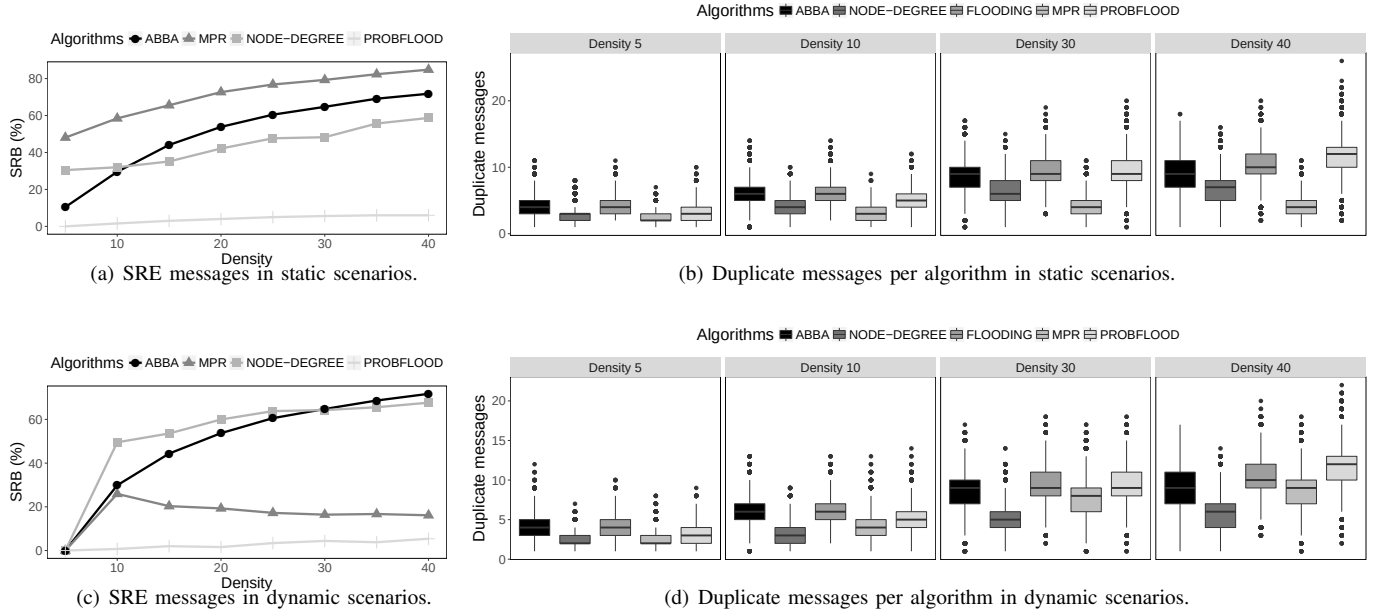


Fig. 2: Overload of wireless communication through *SRE* and *duplicate* messages.

in a *pure flooding* scenario one can expect a distribution where all nodes receive around 30 duplicate messages. However, as shown in Figure 2(b), most nodes barely receive 10 duplicate messages. In general, there are three reasons to get a low number of duplicate messages: i) a higher value of SRE; ii) the occurrence of collisions; and iii) broadcast messages reach few peers (low reachability).

On the one hand, the topology based technique MPR is again able to cope with all densities in both scenarios of mobility. On the other hand, *node-degree* behaves similarly to *ProbFlood* for low densities; but it outperforms *ProbFlood* for density 20 and beyond. *ProbFlood*'s behavior is specially poor in this metric, it is as bad as *simple flooding* when the density is high.

B. Power Consumption

Power consumption of nodes depends, for instance, on the number of messages that nodes exchange or on the amount of time nodes' transceivers spend in their operational modes. The number of received and sent messages are important because, every time a node sends or receives a message, the *transmitter* and/or *receiver* components of the radio change to a more power hungry mode. Hence, the broadcast messages retransmitted as well as the control messages have a direct influence on energy consumption. The *network saturation in the vicinity* of the node is also important due to the way the MAC layer is implemented.

In CSMA, a node must receive an *ACK* notification when a message is received, otherwise the message is sent again. The goal of this strategy is to increase the reliability of the network, but it has a strong negative effect in terms of energy consumption when the network is saturated. Since in this case both messages and *ACK* notifications are often lost, the MAC

layer keeps resending message and consuming energy in the process.

Figures 3(a) and 3(b) show the values of energy consumption for the static and mobility settings respectively.

We use again *boxplots* to present the full distribution of power consumption. Due to space limitations, we do not present the plots for all densities. Instead, we are only showing plots for those densities where there is a significant change in the patterns of energy consumption. The first thing to note is that, as expected, *simple flooding* consumes more energy in all scenarios. Also interesting is that both CDS-based approaches, namely *MPR* and *node-degree*, always outperform other approaches when there is no mobility. This is consistent with the results depicted in Figures 2(a) and 2(b), which show that nodes send and receive fewer messages in these approaches. In the case of *ABBA* and *ProbFlood*, our experiments do not show a single value of density where these algorithms behave better than *MPR*. This is interesting because it indicates that properly using control messages to build an underlying topology is worth in terms of power consumption when the interval among broadcast messages is close to the interval among control messages.

As depicted in Figure 3(b), the behavior of these protocols changes in dynamic scenarios. In such scenarios, *ABBA* outperforms *MPR* when the density is low or medium (5, 10, 15). Only when the density is greater than 20, *MPR* becomes competitive again. In correlating these results with Figures 2(c) and 2(d) we notice that nodes send fewer messages with *ABBA* than with *MPR*. We also notice that the number of duplicate messages received using different algorithms in dynamic scenarios is closer than that of the static scenarios. We

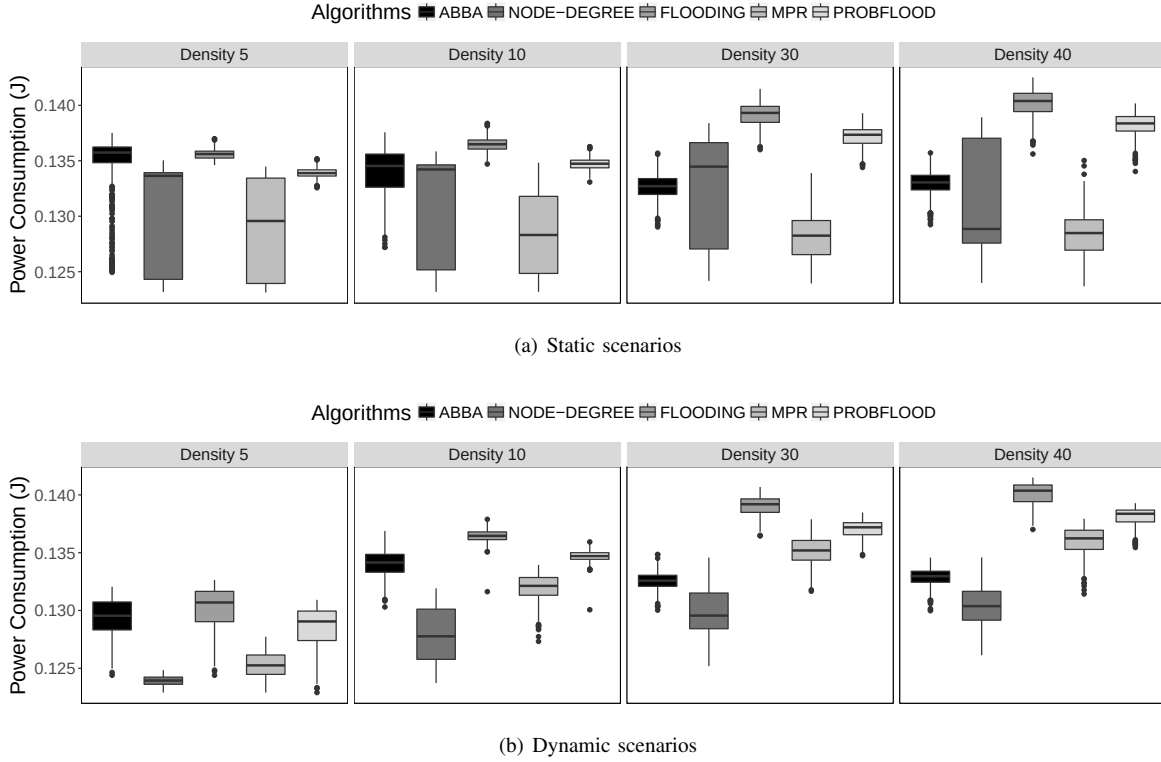


Fig. 3: Power consumption per algorithm

can conclude that, since *MPR* cannot maintain a completely up-to-date underlying topology due to mobility, it fails in reducing the number of relay nodes and the power consumption is high.

These results are of interest because they support the idea that two parameters of a scenario, namely node mobility and ratio *broadcast-interval/control-message-interval*, may impact the decision of which algorithm to use to broadcast.

C. Reachability

Figures 5(a) and 5(b) depict the reachability of the five algorithms studied, in both *static* and *dynamic* scenarios. All these algorithms tend to be reliable in static settings and for densities above 10. Some approaches fail to deliver all broadcast messages in settings with very low density because many sections of the network are only connected by a single link that can be a point of failure. In particular, *ABBA* and *MPR* are more prone to this kind of situations.

The results in the mobility scenarios are less stable. As expected, some protocols such as *MPR* and *node-degree* behave worse because the underlying topology they build to drive the communication must be constantly updated due to the movement of nodes. This problem can be observed for all the densities, but it is more clear for denser topologies. We can also see that *ABBA* tends to fail with different values of density, although far less than the CDS-based approaches. This is understandable because *ABBA* is an area-based approach where each node uses the location of its neighbors to drive

the broadcast process. Since the location of each node changes continuously, it is expected that this approach will not reach a 100% reliably. This is consistent with previous studies [9], [10].

VI. DISCUSSION

Table V summarizes the main results of our experiments. It has the form of a set of rules that allows us to choose a good broadcast algorithm for a given deployment scenario. For instance, when there is no mobility, it is better to use *MPR* no matter the density nor the frequency of messages because, as our experiments show, in such scenarios *MPR* consumes less energy, guarantees good reachability and uses efficiently the medium.

Mobility	Density	Suggestion	Advices
Static	Any	<i>MPR</i>	Reduce frequency of control messages Best coverage
Walking Speed	5	<i>simple flooding</i>	
Walking Speed	10	<i>MPR</i>	
Walking Speed	15+	<i>node-degree</i>	

TABLE V: Rules on which broadcast algorithm to use in different deployment scenarios.

The rules for deployment scenarios with mobility at walking speed are more complex, as they take into account other external parameters such as the density. To suggest an algorithm, we first aim at increasing the coverage, then the energy consumption

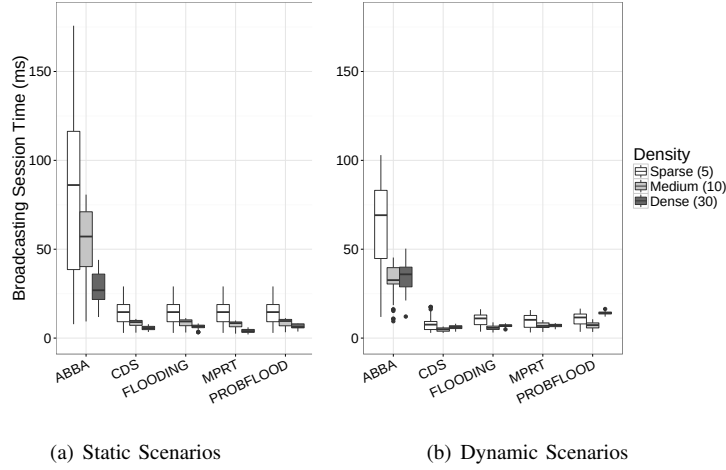


Fig. 4: Broadcasting session time

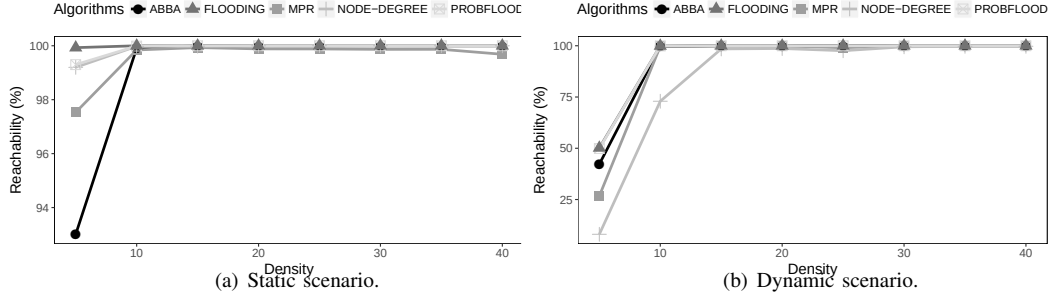


Fig. 5: Node reachability

and we consider that the medium usage is the least important of the metrics. For that reason, we propose *simple flooding* when the density is very low.

Finally, the results show that *ABBA* would be a good alternative when there is mobility and when the density is above 10. Although for walking speed *ABBA* is outperformed by CDS-based protocols, it is worth considering it as the main alternative for scenarios where there nodes move at higher speeds.

VII. PERSPECTIVES

The experiments presented in Section V and discussed in the previous section show that external parameters such as mobility, density, and the interval between broadcast messages have a strong impact on the actual performance of broadcast algorithms. Interestingly, all these parameters can be monitored at runtime. For instance, nodes in the network can detect the density in its vicinity by sending periodic *HELLO* messages or by listening to the traffic. In the same way, many modern mobile devices are equipped with positioning systems and/or accelerometers that can be used to sense the movement of nodes.

Our next step is using the results of this paper to guide the design of an adaptive overlay for disseminating messages in MANETs. In our vision, a monitoring process in each node provides data to a management layer that, in a distributed way, decides what basic approach a node should use to disseminate messages (e.g., *MPR*, *ABBA*, or *simple flooding*). The main challenge in this case is replacing the algorithm used at runtime without losing any message. Fortunately, most dissemination protocols do not require keeping a complex state on each node. Hence, we think that they are interchangeable while keeping strong guarantees on message delivery and broadcast correctness, and that it is possible to have, for instance, an implementation of *ABBA* interacting with an implementation of *simple flooding* or *MPR* through simple translation layers.

VIII. CONCLUSION

Selecting which is the appropriate broadcast algorithm for a given MANET deployment scenario is a daunting task. The performance of all existing approaches strongly depends on this deployment context. The context makes hard to find an adequate approach with the right balance between reliability and efficiency.

The main contribution of this work is to provide a comprehensive comparison of effectiveness of broadcast algorithms for MANETs that helps to take an informed choice between them based on such target deployment scenarios.

Our study reveals that the best algorithm for a given situation, such as a static network, is not the most appropriate for other situations where there is for instance mobility. More specifically, we have compared in our work five representative algorithms in different scenarios that we describe in terms of network density and node mobility. The results obtained allow us to identify what characteristics of these approaches are correlated to a certain performance outcome. Using this information, we can identify what are the algorithms that better match the studied deployment scenarios.

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